1. Definition of Forward Chaining

Forward chaining is a **data-driven inference method** used in Artificial Intelligence.

It:

- Starts from known facts
- Applies logical rules
- Derives new facts
- Continues until no new facts can be generated or goal is reached

It is commonly used in:

- Expert systems
- Rule-based AI systems
- Knowledge-based systems

2. Simple Example Problem

Knowledge Base (Rules):

R1: A → B
R2: B → C
R3: C → D

Initial Fact:
A is True

Expected Output:
D should become True after forward chaining.

3. Python Code for Forward Chaining

```
# Simple Forward Chaining Implementation

# Step 1: Define initial facts
facts = {"A"}   # A is initially true

# Step 2: Define rules as (premise, conclusion)
rules = [
    ("A", "B"),
    ("B", "C"),
    ("C", "D")
]

# Step 3: Forward Chaining Algorithm
changed = True

while changed:
    changed = False
    for premise, conclusion in rules:
        # If premise is true and conclusion not already known
        if premise in facts and conclusion not in facts:
            print(f"Rule applied: {premise} → {conclusion}")
            facts.add(conclusion)
            changed = True

# Step 4: Final Facts
print("\nFinal facts derived:")
print(facts)
```

4. Explanation of the Code

1. We store known facts in a set:
   facts = {"A"}
2. Rules are stored as tuples:
   ("A", "B") means A → B
3. The algorithm:
   - Loops through rules
   - If premise is in facts
   - Adds conclusion to facts
   - Repeats until no new facts are added
4. Output will be:

```
Rule applied: A → B
Rule applied: B → C
Rule applied: C → D

Final facts derived:
{'A', 'B', 'C', 'D'}
```

5. How This Demonstrates Forward Chaining

Step-by-step reasoning:

A is true
→ B becomes true
→ C becomes true
→ D becomes true

The system moves forward from facts to conclusions.

6. Classroom Enhancement Idea (For Your AI Students)

You can modify the example:

Rules:
Fever ∧ Cough → Flu
Flu → Rest

Extend code to support multiple conditions in premise.